CS 6515/4540 (Fall 2025)

Exam 1

Name: _____

GTID: _____

GT Username: _____

INSTRUCTIONS

1. When the exam starts, write your name (or GT ID or GT Username) on each page.
2. Bringing two pages of notes is permitted.
3. No books, calculators, or other electronic devices permitted.
4. Make your answers clear and concise.
5. This examination contains **9 (nine)** pages. Verify no pages are missing.
6. ONLY write on the **FRONT** of each sheet. Backs will not be scanned.
7. You may refer to standard algorithms from class without giving full pseudocode.

I am aware of and in accordance with the Academic Honor Code of Georgia Tech and the Georgia Tech Code of Conduct. I will use no person's help on this test. Also, I have read all the instructions on this page.

Signature:

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1 Short Answer Questions (35 points)

Unless explicitly mentioned, a short 1–2 line justification is required.

1. (No justification required.) What is the smallest odd number M for which running the Median-of-Medians algorithm with groups of M numbers still yields a linear runtime?

2. Given $n \geq 4$, we test primality by checking divisibility for all integers $2 \leq i \leq \lfloor \sqrt{n} \rfloor$. This is a polynomial-time algorithm (in the input size).

   ◯ True

   ◯ False

3. Suppose $k = \log n$. Then $k^k = \Theta(n)$.

   ◯ True

   ◯ False

4. Let M and M' be two matchings in a bipartite graph G. Consider the symmetric difference $\Delta = M \oplus M'$. Then, Δ may contain a cycle of length 7.

   ◯ True

   ◯ False

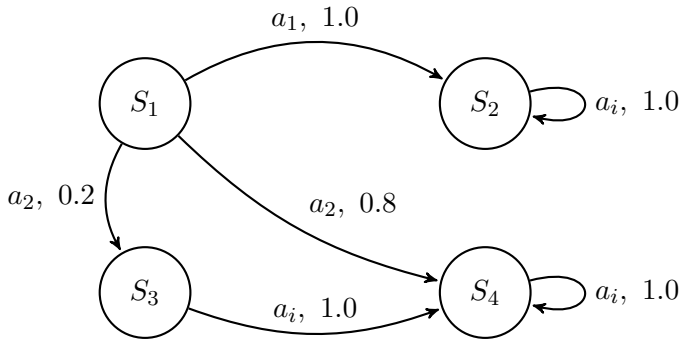
5. Given a weighted graph $G = (V, E)$, let $e = (u, v)$ be the lightest edge incident to u. Then e is in *some* minimum spanning tree of G.

   ◯ True

   ◯ False

6. Consider the MDP with 4 states $\{S_1, S_2, S_3, S_4\}$ and 2 actions $\{a_1, a_2\}$ as given in the figure. Here, the tuple a, p on an edge means that if we perform action $a \in \{a_1, a_2\}$ at the start state then we end at the destination state with probability p. (When we write a_i in the figure that means this is true for both a_1 and a_2.)

   Moreover, we receive a reward 0 on pulling any action in S_1 or S_4; we receive a reward 10 on pulling any action in S_2; we recieve a reward 100 on pulling any action in S_3. Starting at S_1, what is the optimal policy for horizon $T = 2$? Justify briefly.



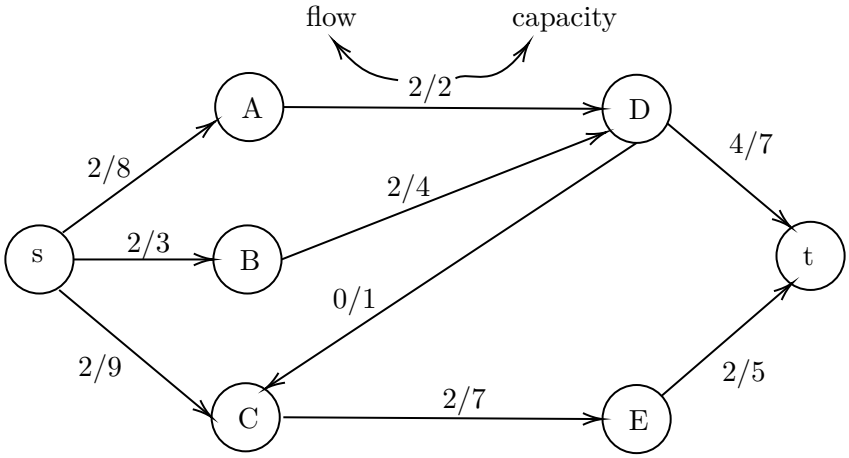
7. A PTAS achieves a $(1 - \epsilon)$-approximation in time polynomial in the input size and in $1/\epsilon$.

   ◯ True

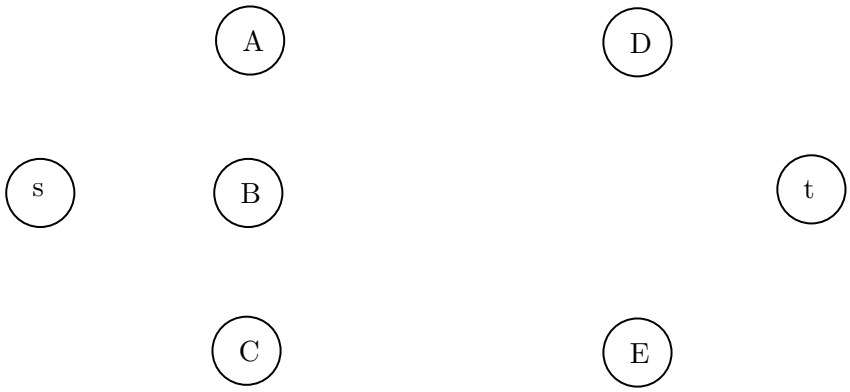
   ◯ False

2 Graphs (20 points)

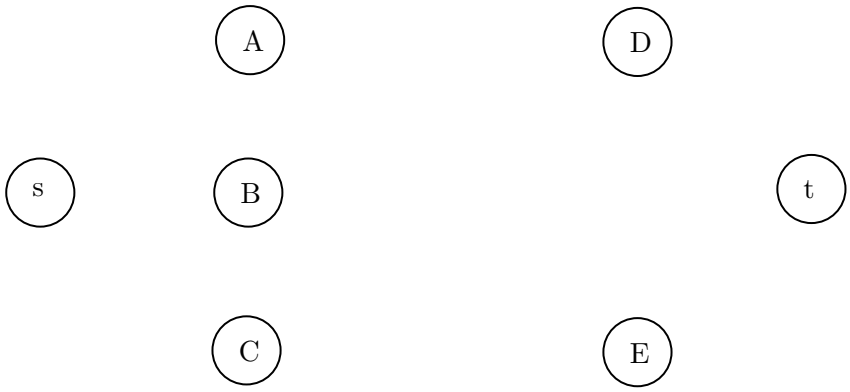
Consider the following st-flow f in a directed graph G, where a/b on an edge denotes flow a and capacity b.



1. (6 pts) Draw the residual graph G^f. Use the convention that for each ordered pair (u, v) you show the residual capacity.



2. (3+3 pts) Find an augmenting path P (or state that none exists) that sends at least 3 units of flow. If P exists, send 3 units of flow along P and draw the resulting flow f'.



3. (4 pts) What is the value of the maximum st-flow in G?

4. (4 pts) List the edges of a minimum st-cut of G.

3 Divide and Conquer (25 points)

Given an array $A[0], A[1], \ldots, A[n-1]$, count the number of inversions, i.e., pairs (i, j) with $i < j$ and $A[i] > A[j]$, in $O(n \log n)$ time.

1. (5 pts) Design `MergeAndCount`(X, Y) that takes two **sorted** arrays and returns (i) the merged sorted array and (ii) the number of pairs (i, j) with $X[i] > Y[j]$.
   Example: `MergeAndCount`([1,5],[2,3]) $\rightarrow$ [1,2,3,5] and 2.

2. (10 pts) Using `MergeAndCount` (assume it is implemented), design a divide-and-conquer algorithm to count inversions in A. Give a high-level description of the recursion.

3. (10 pts) Assume $n = 2^k$ and the runtime satisfies

$$T(1) = 1, \qquad T(n) = 2\,T(n/2) + n \quad (n \geq 2).$$

Prove by induction (do not use Master Theorem) that $T(n) = O(n \log n)$.

4 Dynamic Programming (20 points)

You are given a knapsack of budget $B \in \mathbb{Z}_{>0}$ and n items, where item i has size $s_i \in \{1, \ldots, B\}$ and value $v_i \in \mathbb{Z}_{>0}$. Choose $I \subseteq \{1, \ldots, n\}$ with $\sum_{i \in I} s_i \leq B$ to maximize $\sum_{i \in I} v_i$ (you can choose each item only once).

Design a DP whose running time depends only on n and B (and on $\log v_i$ terms due to bit complexity, which we will ignore for ease of calculations).

1. (5 pts) Define your DP table entries $T(i, b)$, where $i \in \{1, \ldots, n\}$ and $b \in \{1, \ldots, B\}$.

2. (8 pts) Give the recurrence in terms of smaller subproblems and briefly justify correctness.

3. (7 pts) Analyze runtime in terms of n and B. Is the algorithm polynomial-time (in the input size)?

END OF EXAM